

More than Task Performance: Developing New Criteria for Successful Human-AI Teaming Using the Cooperative Card Game Hanabi

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ABSTRACT

As we shift to designing AI agents as teammates rather than tools, the social aspects of human-AI interaction become more pronounced. Consequently, to develop agents that are able to navigate the social dynamics that accompany cooperative teamwork, evaluation criteria that refer only to objective task performance will not be sufficient. We propose perceived cooperativity and teaming perception as subjective metrics for investigating successful human-AI teaming. Corresponding questionnaire scales were developed and tested in a pilot study employing the collaborative card game Hanabi, which has been identified as a unique setting for investigating human-AI teaming. Preliminary descriptive results suggest that rule-based and reinforcement learning-based agents differ in terms of perceived cooperativity and teaming perception. Future work will extend the results in a large user study to psychometrically evaluate the scales and test a conceptual framework that includes further aspects related to social dynamics in human-AI teaming.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**; **HCI theory, concepts and models**; **Empirical studies in HCI**; **User models**; **User studies**; **Laboratory experiments**; **Collaborative and social computing design and evaluation methods**; **Collaborative interaction**.

KEYWORDS

Human-AI teaming, human-autonomy teaming, cooperation, collaboration, social perception

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1 INTRODUCTION

In recent years, disruptive advances in the field of artificial intelligence (AI) have emerged and entered commercial applications. Conversational agents (CAs) based on Large Language Models, such as ChatGPT [8] or Google Bard [31] have become increasingly integrated into the daily lives of people, both in professional and private areas [19]. CAs feature characteristics of human social interaction through the use of generative AI for natural language processing, which paves the way for large-scale changes in human-AI interaction. Moreover, the general paradigm shift of designing autonomous agents as teammates rather than tools [6, 23, 41] will contribute to an amplification of social dynamics within human-AI interaction [28]. As a consequence, AI developers have to face new challenges, as technology will only be optimally adopted if it is able to navigate the social dynamics that accompany cooperative teamwork [15]. Thus, it has been claimed that developing agents with teaming intelligence will be the next big frontier in AI research [4, 15, 35, 45]. Along these lines, the question arises as to how interaction success within human-AI teaming is to be understood and, thus, which optimization criteria should be applied when designing AI along social interactions. Following [3, 21, 28, 33, 43] we argue that interaction success in human-AI teaming is determined not only by objective task performance but also by specific subjective facets of user experience. While previous work on human-centered AI [12] has highlighted the crucial role of team cognition (see [41]), for instance regarding (shared) mental models [2, 3] and perceived mutual understanding [9], we focus on aspects related to social processes involved in human-AI teaming [2], namely (1) the perception of the AI agent(s) as being cooperative and (2) the perception of the parties as a team [6, 21, 28, 47]. However, the theory-driven conceptualization, operationalization, and empirical investigation

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of subjective human-AI teaming success criteria related to social processes are still in their infancy [6, 28, 30].

The goal of the present work is to address this gap by proposing *perceived cooperativity* and *teaming perception* as two core variables that may shape human-AI teaming for which, to our knowledge, no standardized measures exist up to now. Based on joint activity theory [7, 22] and research on group dynamics (e.g., [49]) we developed questionnaire scales as a first step to make these two variables quantifiable. We tested both scales in a pilot study in which participants played an online version of the collaborative card game Hanabi, paired with artificial agents and human players. Hanabi has been identified as a unique study setting for investigating both human-human and human-AI teaming [27, 42, 46] due to its inherent demand of collaborative behavior among players, which requires social cognitive skills [26] and information exchange using restricted communicative actions [16].

In a large user study with a planned sample size of $N = 180$ we will expand the present work by employing the scales within the Hanabi game setting to enable advanced statistical methods to analyze scale reliability, dimensionality, and validity, as well as to explore relationships between perceived cooperativity, teaming perception, and objective game metrics (e.g., synergy [51, 52]). Moreover, links to established factors shaping social dynamics in human groups (i.e., based on the stereotype content model [17, 29, 32]), human-AI interaction (subjective information processing awareness [44]), and user diversity (trait altruism [13], affinity for technology interaction [18]) will be examined.

2 BACKGROUND

2.1 Human-AI Teaming from a Psychological Perspective

To define success criteria for human-AI teaming, it is crucial to elucidate the various terms encompassing the spectrum of engagements with AI systems [6]. According to joint activity theory [22] interaction between humans and AI can comprise independent or interdependent, joint activity. Further, joint activity can be subdivided into cooperation and collaboration, depending on which elements of the activity are interdependent: if only the actions are interdependent, the joint activity is a cooperation; if the parties also have shared goals, the joint activity is a collaboration [22]. Lastly, the term teaming refers to collaboration explicitly involving aspects of social bonding (e.g., team cohesion, parties having team roles [6, 41]).

Besides interdependency of actions, joint activity theory [22] specifies another prerequisite for joint activities to emerge: All parties (i.e., humans or artificial agents) have to agree to participate in the activity, which necessitates some degree of goal alignment. This can include setting aside short-term goals in order to achieve long-term goals (see also [48]). For joint activities (i.e., cooperation, collaboration, and teaming) to be successful, three requirements have to be fulfilled: The parties have to (1) create a common ground (i.e., manage mutual knowledge, beliefs, and assumptions), (2) be mutually predictable, and (3) be mutually directable [7, 22]. These requirements can be understood, on an individual level, as the ability and willingness to contribute to a successful joint activity—which we want to refer to as *cooperativity*. Moreover, these requirements

can be subjectively perceived by interaction partners. Hence, they appear fruitful as a source for assessing the subjective perception of partner cooperativity, or: *perceived cooperativity* (PC).

One aspect of human-AI teaming that remains understudied so far [6] is group dynamics (e.g., shared situation awareness, team cohesion, groupthink [1, 6, 25]). Central here is the foundational question: Is it conceivable for humans and artificial agents to consolidate into a cohesive team, wherein these dynamics emerge [1]? Findings focusing on group dynamics within human-machine teams have shown that, indeed, team performance and user satisfaction increased if the machine was perceived as a teammate [37, 49]. However, building on the notion that joint activity requires *mutual* predictability and directability [22], a cohesive team also comprises the perception of *oneself* as a teammate, i.e., the perceived own contribution to achieving the partner's or shared goals (e.g., [14]). Hence, only investigating in how far the AI agent is perceived as cooperative is not sufficient. Consequently, we aimed at developing a scale to assess the subjective perception of (1) oneself and of (2) the partner as being contributors to shared goals and mutual supporters as well as (3) the team effort as a whole. In the following, these three aspects will be termed *teaming perception* (TP).

2.2 Hanabi as a Unique Setting for Human-AI Teaming Research

Similar to games like Chess or Go, Hanabi has been proposed as a benchmark for developing machine learning algorithms [4], with the difference that Hanabi serves the next big challenge in AI [15]: It inherently demands collaborative play and reasoning about a partner's mental state, which is managed through restricted communicative actions [16] (for a depiction of Hanabi rules see Appendix A). As such it also offers a rich embedding for investigating human-AI teaming, since it fulfills the two requirements that have been said to be necessary to establish a collaborative joint activity [7, 22]: When entering the game, both players implicitly agree to collaborate towards a shared goal (i.e., scoring a maximum of points) through interdependent actions (i.e., game moves that are interlinked between players). These two requirements are inherent, non-negotiable characteristics of the Hanabi game.

Furthermore, to succeed in Hanabi, players have to establish and manage common knowledge by sharing exclusive information and making assumptions about the other players' beliefs regarding the current state of the game [4]. Mutual predictability can be achieved by displaying a reliable and comprehensible game style. Mutual directability is possible through (1) explicit hints and (2) establishing implicit communication codes (i.e., "hidden rules", which have been found to make human players superior to AI agents up to now [27, 46]). Hence, although Hanabi requires collaborative play (i.e., players have no choice but to collaborate if they want to play), the degree of player *cooperativity* is variable, making it especially valuable for investigating subjective factors linked to successful human-AI teaming. Moreover, due to Hanabi's restricted set of communicative actions (i.e., game moves, response latency) particularly when played digitally, player behavior can be measured, analyzed, and manipulated in a highly standardized manner. This also indicates that digital implementations of Hanabi can be used to make human-human and human-AI teaming easily comparable.

So far, however, Hanabi has been rarely utilized to examine psychological mechanisms underlying human-AI interaction [39, 47]. In [47], participants played with one rule-based and one reinforcement learning-based Hanabi agent. It was found that while task performance (i.e., game score) was similar for both agents, subjective user experience was more positive after playing with the rule-based agent, but only for experienced Hanabi players. In [39] agents imitating cognitive processing were associated with higher subjective ratings of human-likeness, likability, and intelligence. Both studies did not include a baseline with human players collaborating. Moreover, the studies did not adopt an explicit human-centered, experiential perspective in human-AI teaming [43], restricting conclusions as to what the antecedents of successful human-AI teaming in terms of user experience factors are. In the present work, we build on this past research by testing our two new developed scales for estimating PC and TP when collaborating with other human players as well as one rule-based and one reinforcement learning agent playing Hanabi.

3 METHOD

3.1 Pilot Study Design and Procedure

A 2x2x2 mixed experimental design was used. As independent variables we varied partner type (human vs. artificial; within subjects), artificial agent type (Piers vs. RL, see Section 3.3.2; within subjects), and artificial agent response latency (fixed vs. random; between subjects; details regarding the latency manipulation will not be reported here). We considered the human partner (referred to as Human-Human, short: HH) condition as an empirical baseline for comparing the artificial agents (referred to as Human-Agent, short: HA) condition to. Ethics approval for this study was obtained from the Ethics Committees of University of Lübeck (Tracking number: 2023-570) and HRI Europe.

As subjective dependent variables we assessed PC, TP, playing style similarity (PSS), social perception [17, 29, 32], and subjective information processing awareness (SIPA [44]). Since the type of game partner was not disclosed before the games (i.e., participants were unaware if they played with a human or artificial game partner), we assessed the participants' estimation of the player type (human partner confidence, HPC; human game partner experience; HPGE). Finally, we asked participants if they would like to switch their game partner and why. As objective dependent variables we assessed game performance metrics (i.e., score, number and distribution of moves, number of errors). Results regarding social perception, SIPA, and game partner switch will not be reported here.

The pilot study was separated into three parts: (1) A pre-questionnaire assessing control variables, (2) a game session of two rounds of Hanabi, paired with one other human player (HH), and (3) a game session of four rounds of Hanabi paired with artificial agents (HA), of which two rounds were played with a rule-based agent (Piers [50]) and two rounds with a reinforcement learning agent (RL [11]). After both rounds were concluded for each experimental condition, participants completed questionnaires for assessing the dependent variables. The pre-questionnaire was completed in approx. 5 min., the HH session in approx. 60 min., and the HA session in approx. 90

min. The order of the sessions and artificial agents within sessions was balanced.

3.2 Participants

Eight participants were recruited for the pilot study at HRI Europe. Their mean age was $M = 42.00$ ($SD = 9.04$); four were women, four were men. Three participants reported to have prior Hanabi experience. Participants' mean affinity for technology interaction (ATI) was $M = 4.04$ ($SD = 1.12$) on a possible score range of 1–6; mean trait altruism was $M = 3.97$ ($SD = 0.70$) on a possible score range of 1–5.

3.3 Study Setup

3.3.1 Hanabi Online Setup. To allow participants to play Hanabi in a standardized setting, we developed an online setup based on the *Hanabi Learning environment* by [4] (available from <https://github.com/google-deepmind/hanabi-learning-environment>) and a custom web-based user interface (Figure 1, available from <https://github.com/HRI-EU/Hanabi4HMI>). Within each condition, participants played two rounds of Hanabi with two fixed random initializations (Appendix B).

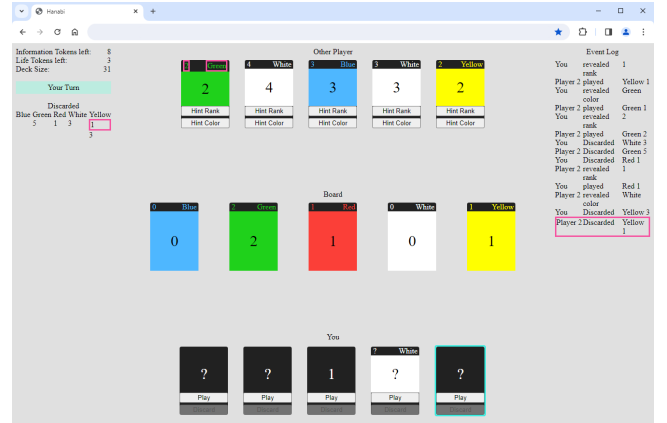


Figure 1: User interface of the Hanabi online setup used.

3.3.2 Artificial Hanabi Agents. Two types of artificial agents were used in the study: a rule-based agent (Piers) introduced by [50] and an RL agent as described and trained in [4, 11]. We used implementations of both agents presented in [11] (available from <https://github.com/rocanaan/hanabi-ad-hoc-learning>). These agents were chosen based on their high performance in benchmark tests [10] (see Appendix C). Moreover both agents were designed to mimic human behavior and found to play well with human partners in ad-hoc team settings, with RL agents typically outperforming rule-based agents [10, 11].

The rule-based Piers agent determines its next move based on an ordered set of rules (see Appendix D) that are modeled after human game play and are designed to generate cautious behavior that aims at never losing a life token.

The RL agent uses a Rainbow deep Q-network (DQN) architecture [4, 20, 36], comprising of a 2-layer Multi-Layer Perceptron

of 512 hidden units each to approximate value distributions using distributional reinforcement learning [5]. As input, the network receives a binary vector with 658 entries representing the current game state observable by the agent and the most recent action. The agent's action is selected ϵ -greedily over the expected value of each of the 20 possible actions (for details and hyperparameter settings see [4, 10]). During training, the RL agent played the *Piers* agent as partner. Furthermore, the followed training procedure and the non-recurrent RL architecture used here are assumed to lead to policies that rely on immediately available game information instead of long-term strategies and complex conventions. Thus, these policies are more likely to adapt well to previously unseen agents and strategies [10].

3.4 Subjective Measures

In the following, we present a broad overview of the scale development for PC and TP. Further, we describe items used for partner type estimation and playing style similarity (PSS), as well as standardized scales for ATI and trait altruism. If not otherwise stated, participants responded to the questionnaire items on a 6-point Likert scale from 1 (*completely disagree*) to 6 (*completely agree*). For the self-constructed scales, we chose the 6-point response format to avoid misuse of a neutral middle category [24]. Due to the small sample size of the pilot study, we refrained from reliability analyses. These will be subject of the planned study with $N = 180$.

3.4.1 Perceived Cooperativity. Since we define PC as the subjective perception of the partner's ability and willingness to contribute to a joint activity, the requirements for successful joint activities as listed in the summary of [7] formed the basis of our item design. Here, the authors name eight broad requirements (i.e., a good agent should be observable, aware of the task progress, informative, aware of its limitations, predictable, dependable, directable, and selective). We disaggregated these into single aspects of cooperative partner behavior (i.e., an item should encompass only one aspect to be evaluated) and translated them into a questionnaire item format. We also adapted the scale to the specific Hanabi context by altering the wording of three items. The resulting scale comprised 15 items (see Table 3 for the human player and Table 4 for the artificial agent version). Please note that this scale version is subject to further development, hence, we refer to the scale reported here as Version 1 (V1).

As this was the first deployment of the PC scale, we additionally aimed at testing if the items were contextually appropriate (i.e., whether participants felt confident in rating the partner on the respective items based on the interaction in the Hanabi setting). To this end, participants also rated their confidence in the item response on a 11-point rating scale from 0 (*not sure at all*) to 10 (*absolutely sure*).

3.4.2 Teaming Perception. We define TP as the subjective perception of the entire group as a team, which includes the perception of the partner and oneself as teammates. We therefore developed items that relate to both the group as a whole and its individual members (similar to [37, 47]). In line with work on the evolutionary origins of cooperation within human social groups (e.g., [34, 48]), item development focused on collaborative effort, mutual support,

own contribution to the team's goal, and setting aside own goals. The resulting scale comprised 9 items (see Table 5 for the human player and Table 6 for the artificial agent version). As with PC, this scale version is subject to further development, hence, we refer to the scale reported here as Version 1 (V1). Moreover, as with PC, participants rated their confidence in the item response on a 11-point rating scale from 0 (*not sure at all*) to 10 (*absolutely sure*).

3.4.3 Partner Type Estimation and Playing Style Similarity. To assess partner type estimation and PSS, self-constructed items were used (see Table 7). Partner type estimation was subdivided into human partner confidence (HPC) and human partner game experience (HPGE).

3.4.4 Control Variables. For assessing prior Hanabi experience, we asked participants whether they have played Hanabi before (yes/no). If yes, they were asked to estimate their prior experience on a rating scale ranging from 1 (*I played Hanabi only once or twice*) to 6 (*I would consider myself a Hanabi expert*). For assessing affinity for technology interaction (ATI), the 9-item ATI scale [18] was used. For assessing trait altruism, the 8-item altruism subscale of the German version of the revised NEO personality inventory [40] was used. Responses were provided on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Assessed demographics were age and gender.

3.5 Game Data

The Hanabi Learning environment allows for automatic tracking the following game data: player type, game score, type of every single move, latency of every single move, and the current game state including each players' observations and card knowledge, current fireworks, life and information tokens, deck size, and discarded cards. From these data, we calculated number of errors by partner as the number of erroneously played cards that led to the loss of a life token, and the total game score as the sum of the maximum card rank over the five fireworks stacks. To allow for first insights into the possible effects of player type and agent type on the objective task performance, we report results on game score, distribution of game moves, and number of errors.

4 RESULTS

Since the present study was a pilot study with a very small sample size, we report descriptive data only. Inferential statistical methods for item reliability, dimensionality, and validity analyses as well as hypotheses testing will be employed in the planned study.

4.1 Subjective Data

As a first step, descriptive statistics were calculated for PC single items as well as their associated confidence ratings (see Table 8 for HH and Table 9 for HA). Confidence ratings in the HH condition were uniformly above the scale midpoint 5.0 (mean confidence rating: $M = 6.77$, $SD = 1.25$), while one confidence rating in the HA conditions was below the scale midpoint 5.0 (Piers: Item PCA12). Mean confidence ratings in the HA conditions were slightly lower than in the HH condition (Piers: $M = 6.01$, $SD = 1.16$; RL: $M = 6.50$, $SD = 1.20$). The same was repeated for TP (see Table 10 for HH and Table 11 for HA), where confidence ratings in all conditions

were above the scale midpoint 5.0 (HH: $M = 7.51$, $SD = 1.66$; Piers: $M = 6.56$, $SD = 1.32$; RL: $M = 6.85$, $SD = 1.23$). For both PC and TP, average scores in the HH condition were higher than in the HA condition, and average scores for the Piers agent were higher than for the RL agent (Table 1). The largest difference between the two artificial agents in average score for a single item was found for PCA10 (“I had the feeling that the other player was dependable.”) and TPA07 (“The agent contributed to reaching the shared goal.”), in the same direction as the overall results.

Regarding partner type estimation, the score values for the human partner and the Piers agent were similar, both for HPC and HPGE, while values for the RL agent were lower. Differences in PSS could be observed between all conditions, with highest score values for human partners, followed by the Piers agent and lastly the RL agent.

4.2 Game Data

We excluded all games that scored less than 5 points in total from the analyses. We did however include games that were lost at a later stage due to the loss of all life tokens. We considered the total score when losing the last token for analysis. In the HH condition, 2 games were lost, while 4 games with the Piers agent were lost, and 8 games with the RL agent. Games in the HH condition ended with a higher mean score ($M = 18.9$, $SD = 3.0$, Figure 2A) than in the HA conditions (Piers: $M = 16.4$, $SD = 3.9$; RL: $M = 13.3$, $SD = 4.6$). The average game length measured by the number of moves was similar in games with human players ($M = 65.1$, $SD = 13.3$, Figure 2B) and with the Piers agent ($M = 66.5$, $SD = 12.3$), while games with the RL agent were shorter ($M = 57.3$, $SD = 12.1$). The total duration of a game measured in s was highly correlated with the game score and is thus not reported. The mean number of errors was similar in games with human players ($M = 1.8$, $SD = 0.9$, Figure 2C) and with the Piers agent ($M = 1.9$, $SD = 1.0$), while it was higher in games with the RL agent ($M = 2.6$, $SD = 0.7$). In the HH condition, participants more often played a card ($M = 0.32$, $SD = 0.08$, Figure 2D) than in the HA conditions (Piers: $M = 0.20$, $SD = 0.07$; RL: $M = 0.22$, $SD = 0.07$). Conversely, the relative frequency of hints (rank or color) was lower in the HH condition ($M = 0.40$, $SD = 0.08$) compared to the HA conditions (Piers: $M = 0.53$, $SD = 0.12$; RL: $M = 0.54$, $SD = 0.16$). The frequency of discards was similar (HH: $M = 0.28$, $SD = 0.05$; Piers: $M = 0.27$, $SD = 0.082$; RL: $M = 0.26$, $SD = 0.09$).

5 DISCUSSION

The objective of the present work is to conceptualize and operationalize *perceived cooperativity* (PC) and *teaming perception* (TP) as novel subjective metrics to investigate and evaluate the underlying dimensions of successful human-AI teaming from an individual user perspective. To that end, we propose two scales that were theoretically based on the work of [7, 22], defining requirements for successful joint activity in the HMI context. We implemented the scales in a first pilot study to assess participants’ experience when playing the collaborative card game Hanabi in different experimental conditions. For this purpose, we developed a novel online game setup around the environment by [4] to investigate and compare human-human and human-AI interaction in a standardized way.

Based on the first empirical results we report here, it appears promising that the proposed scales for PC and TP might be able to differentiate between game partners with different levels of teaming intelligence within the Hanabi study setting, thereby enabling deeper insights into the underlying psychological factors that contribute to the perception of successful human-AI teaming. Participants’ ratings for PC and TP were higher when playing with another human player than with the artificial agents. When comparing the artificial agents, it became apparent that the RL agent was perceived as less cooperative than the rule-based agent, which is in line with previous findings [47]. However, different from [47], the subjective user experience seemed to coincide with the objective game data. Since the sample size in the pilot study was too small to test for statistical significance and effect sizes of the differences as well as correlations, the results do not allow for testing hypotheses regarding the causal relationships between the independent and dependent variables. For instance, we hypothesize that the agent type (rule-based vs. RL) has an effect on the traceability of the agent behavior [44], which may directly affect perceived predictability and directability, two aspects that are assessed through our PC scale. Our future work will extend the present study to a sample size of $N = 180$ to (1) be able to psychometrically evaluate the proposed scales and (2) investigate the psychological dynamics underlying human-AI teaming by providing and testing a conceptual framework for the subjective experience of teaming success. For this purpose, we plan to consider additional factors, i.e., anthropomorphism [38], social perception based on the stereotype content model [29, 32], and SIPA [44], as well.

Given that the presented work is a work in progress, there are several limitations to emphasize. First, the empirical data is based on a very small sample size and does not allow for statistical analyses which are necessary to evaluate scale reliability, validity, and dimensionality. Consequently, the scales that are provided below represent an intermediate step towards final scale versions, which will be shared after thorough statistical analyses. Second, as the Hanabi game requires collaborative play to achieve a shared goal, negotiation processes to establish a joint activity in the first place cannot be modeled within the environment. Hence, testing the proposed scales in situations where parties can deviate from cooperation or collaboration (e.g., selfish behavior) needs the employment of a different study setting. Third, the current dyadic nature of the game does not allow for testing the scales in larger team constellations. Cooperation and collaboration can be established and investigated in dyads, but to further explore the interplay between team dynamics (e.g., team cohesion, team roles [6]) and our proposed success criteria *perceived cooperation* and *teaming perception*, further research needs to employ larger groups (e.g., multi-agent settings).

Ultimately, we aim at developing general questionnaire scales for PC and TP in human-AI teaming that can add to evaluating and designing successful human-AI teaming from a more holistic point of view. Therefore, future work should address the above mentioned limitations to further test the scales’ applicability to contexts that go beyond the Hanabi setting. Moreover, scale development also needs to address the trade-off between application specificity (and easier answerability) and generalizability (and harder answerability). Human-AI teaming is a broad field with various types of interaction

Table 1: Descriptive Statistics (M (SD)) of the Dependent Variables by Partner Type

Condition	PC ¹	TP ²	HPC ³	HPGE ⁴	PSS ⁵
Human	4.40 (0.54)	4.99 (0.67)	3.81 (1.41)	4.25 (1.28)	4.13 (0.74)
Agent (Piers+RL)	3.01 (1.09)	3.79 (1.06)	2.94 (1.27)	3.19 (1.40)	2.47 (1.41)
Piers	3.58 (0.97)	4.19 (0.72)	3.63 (0.95)	3.88 (0.87)	3.13 (1.16)
RL	2.43 (0.50)	3.38 (0.78)	2.25 (0.85)	2.50 (1.10)	1.81 (0.80)

¹PC: Perceived cooperativity; ²TP: Teaming perception; ³HPC: Human partner confidence; ⁴HPGE: Human partner game experience; ⁵PSS: Playing style similarity

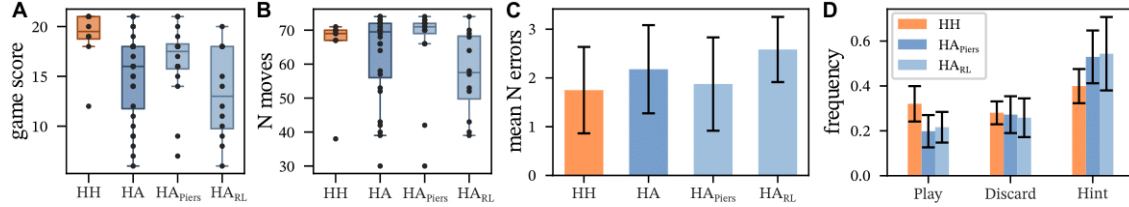


Figure 2: A Box plots showing game score by partner. **B** Number of moves by partner. **C** Mean number of errors by partner (cards played that resulted in loss of a life token). Error bars represent SD . **D** Relative frequency of game moves by partner.

modalities [33], covering—among others—restricted non-verbal interaction as in the Hanabi setting, natural language communication as in CAs, and embodied interaction as in robots. Employing not only the scales for PC and TP but also ratings for participants' confidence in their single item estimations will enable a deeper insight into the scales' applicability as well as their transferability to related applications.

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A HANABI GAME RULES

The rules of the Hanabi card game are as follows. Each player holds five cards that are only visible to the partner(s). Each card has a rank from 1 to 5 and one of five colors. The goal is to play cards such as to complete five “fireworks” stacks (one in each color), on which cards are sorted in sequential, ascending order. Each turn, a player can take one of three possible actions: (1) *Hint*: Tell the partner the rank or color of their cards; a hint means pointing out all cards of one color or rank, (2) *Discard*: Discard a card to the discard pile, (3) *Play*: Play a card.

Players have three life and eight information tokens. Giving a hint costs one token, while discarding a card recovers one token. If a card is wrongly played (i.e., can not be added to an existing fireworks stack or does not start a new one), one life token is lost. The game ends if either all life tokens are lost, no cards remain on the pile, or all five fireworks stacks have been successfully completed. The final game score is the sum of highest card ranks per each of the fireworks stack with the perfect score being 25. For each color, multiple cards are in the game for some of the ranks.

There are three cards of rank 1, two cards or ranks 2, 3, 4, each, and one card of rank 5. This makes erroneously discarding cards of rank 5 especially costly as this immediately makes a perfect score impossible.

B EFFECT OF GAME SEED

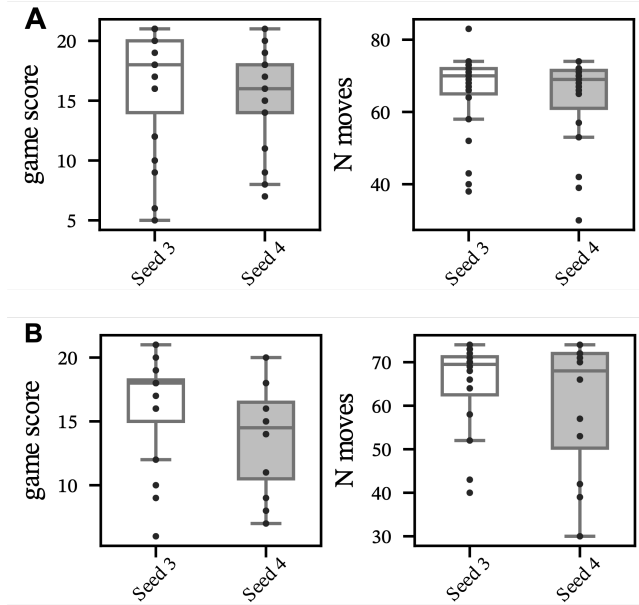


Figure 3: Box plots of game statistics by game seed for A all games, B games in the HA conditions only.

C RESULTS FOR AGENT SELF-PLAY

Table 2: Game Statistics for Agent Self-Play (Rule-based Piers agent and Reinforcement-Learning Agent, RL).

Player 0	Player 1	Seed	Score	N moves
Piers	RL	3	17	73
Piers	RL	4	16	68
Piers	Piers	3	21	72
Piers	Piers	4	15	65
RL	RL	3	5	83
RL	RL	4	17	72

D IMPLEMENTED RULES OF RULE-BASED AGENT PIERS

Ordered from highest to lowest priority, the rules are:

- If the current move is the last move in the game and there is more than one life left, try to score a point by playing the card most probable to score a point
- Play a card that is known to be playable
- If there is more than one life left, play a card that is playable with at least 60 % probability

- Tell partner the remaining unknown suit or rank of a still-playable card
- If less than 4 information tokens are left, tell the partner the remaining unknown information to correctly identify a card as dispensable (if only one hint is needed for identification)
- “OsawaDiscard”: Discard a card that cannot be played at the end of the turn, based on the available knowledge about that card (can be incomplete, e.g., a card with known rank and unknown color will not be playable if all fireworks already have a higher rank on top, or if a card is not playable because all cards required to be played before have been discarded)
- Discard card that has been on the hand the longest
- Give partner a random hint
- Discard a random card

E SELF-CONSTRUCTED QUESTIONNAIRE SCALES

E.1 Perceived Cooperativity Scale

Table 3: Version 1 of the Perceived Cooperativity Scale (Human Player)

Item Code	Item Text: I had the feeling that ...
PCH01	... the other player could communicate their status.
PCH02	... the other player made clear what they wanted to achieve.
PCH03	... the other player could show me how far along they are with their task.
PCH04	... the other player could detect possible difficulties in reaching the goal.
PCH05	... the other player knew enough about my goals to collaborate with me.
PCH06	... the other player could adjust their communication to a concrete situation.
PCH07	... the other player could take or pass the initiative on their own.
PCH08	... the other player could react to my moves.
PCH09	... it was predictable how the other player would react to different moves.
PCH10	... the other player was dependable.
PCH11	... the way the other player played could be influenced by me.
PCH12	... the other player had access to all relevant information.
PCH13	... the other player could point out to the currently most important information.
PCH14	... the other player could detect that they and I had different knowledge for achieving the goal.
PCH15	... the other player shared relevant information with me.

Table 4: Version 1 of the Perceived Cooperativity Scale (Artificial Agent)

Item Code	Item Text: I had the feeling that ...
PCA01	... the agent could communicate its status.
PCA02	... the agent made clear what it wanted to achieve.
PCA03	... the agent could show me how far along it is with its task.
PCA04	... the agent could detect possible difficulties in reaching the goal.
PCA05	... the agent knew enough about my goals to collaborate with me.
PCA06	... the agent could adjust its communication to a concrete situation.
PCA07	... the agent could take or pass the initiative on its own.
PCA08	... the agent could react to my moves.
PCA09	... it was predictable how the agent would react to different moves.
PCA10	... the agent was dependable.
PCA11	... the way the agent played could be influenced by me.
PCA12	... the agent had access to all relevant information.
PCA13	... the agent could point out to the currently most important information.
PCA14	... the agent could detect that it and I had different knowledge for achieving the goal.
PCA15	... the agent shared relevant information with me.

E.2 Teaming Perception Scale

Table 5: Version 1 of the Teaming Perception Scale (Human Player)

Item Code	Item Text
TPH01	The other player and I were a team.
TPH02	The other player and I worked together towards a goal.
TPH03	The other player and I supported each other.
TPH04	I have contributed to reaching the shared goal.
TPH05	I put my own goals aside in favor of the other player.
TPH06	I supported the other player.
TPH07	The other player contributed to reaching the shared goal.
TPH08	The other player put its own goals aside in favor of me.
TPH09	The other player supported me.

Table 6: Version 1 of the Teaming Perception Scale (Artificial Agent)

Item Code	Item Text
TPA01	The agent and I were a team.
TPA02	The agent and I worked together towards a goal.
TPA03	The agent and I supported each other.
TPA04	I have contributed to reaching the shared goal.
TPA05	I put my own goals aside in favor of the agent.
TPA06	I supported the agent.
TPA07	The agent contributed to reaching the shared goal.
TPA08	The agent put its own goals aside in favor of me.
TPA09	The agent supported me.

E.3 Partner Type Estimation, Playing Style Similarity

Table 7: Self-constructed Items to Assess Partner Type Estimation and Playing Style Similarity

Item Code	Item Text
<i>Partner Type Estimation</i>	
HPC01	I am sure to have played with another human.
HPC02	I am sure to have played with an artificial agent.
HPGE01	The game experience felt as if I had played with another human.
HPGE02	The game experience felt as if I had played with an artificial agent.
<i>Playing Style Similarity</i>	
PSS01	The other player/the agent and I had a similar playing style.
PSS02	The other player/the agent and I were very different in the way we played Hanabi.

F ITEM STATISTICS

F.1 Perceived Cooperativity Scale

Table 8: Perceived Cooperativity Scale Item Statistics (V1, Human Player)

Item Code	Item M	Item SD	Confidence M	Confidence SD
PCH01	4.25	0.89	6.13	1.96
PCH02	4.13	0.84	7.00	1.20
PCH03	3.38	1.30	5.63	2.00
PCH04	4.25	0.46	6.25	1.58
PCH05	4.50	0.76	6.63	1.30
PCH06	4.63	1.06	6.38	1.69
PCH07	4.50	0.54	7.13	1.25
PCH08	5.00	0.93	7.38	1.19
PCH09	4.25	1.16	7.38	1.06
PCH10	4.75	0.71	6.63	1.92
PCH11	4.50	0.76	7.50	2.20
PCH12	3.75	1.49	6.13	2.10
PCH13	4.50	0.76	7.13	1.13
PCH14	4.88	1.25	7.13	2.23
PCH15	4.75	0.89	7.13	1.73

Table 9: Perceived Cooperativity Scale Item Statistics (V1, Artificial Agent)

Item Code	Piers				RL			
	Item M	Item SD	Confidence M	Confidence SD	Item M	Item SD	Confidence M	Confidence SD
PCA01	3.13	0.64	5.00	2.07	2.25	1.04	6.13	2.59
PCA02	3.38	1.30	6.25	1.67	2.50	1.07	6.75	1.28
PCA03	2.63	1.51	6.00	2.51	2.38	0.92	5.50	2.39
PCA04	3.13	0.99	5.75	1.39	2.00	0.76	6.88	1.96
PCA05	4.13	0.84	5.50	2.27	3.38	0.74	5.75	2.76
PCA06	3.25	1.16	6.38	1.69	2.25	1.28	6.50	1.60
PCA07	3.88	0.99	5.38	1.85	3.00	1.20	5.25	1.83
PCA08	4.25	1.28	6.75	1.83	2.63	0.74	6.63	1.19
PCA09	4.00	1.07	6.25	1.75	2.38	0.92	7.25	2.19
PCA10	4.13	1.55	6.88	1.64	1.63	0.74	7.75	1.83
PCA11	3.50	1.77	6.75	1.83	2.50	0.76	6.13	1.96
PCA12	3.63	0.52	4.13	2.95	3.00	1.07	6.38	1.30
PCA13	3.38	1.51	7.00	2.00	1.75	1.16	7.63	1.19
PCA14	3.75	1.04	5.63	2.26	2.38	0.92	5.63	2.77
PCA15	3.63	1.30	6.50	1.93	2.50	0.93	7.38	1.77

F.2 Teaming Perception Scale

Table 10: Teaming Perception Scale Item Statistics (V1, Human Player)

Item Code	Item <i>M</i>	Item <i>SD</i>	Confidence <i>M</i>	Confidence <i>SD</i>
TPH01	5.25	0.71	7.63	2.13
TPH02	5.50	0.76	8.13	1.73
TPH03	5.13	0.84	8.25	1.75
TPH04	5.38	0.52	8.00	1.93
TPH05	4.25	1.58	6.50	2.98
TPH06	5.25	0.89	7.63	1.77
TPH07	5.38	0.92	8.00	1.85
TPH08	4.00	1.60	5.88	2.36
TPH09	4.75	0.71	7.63	1.41

Table 11: Teaming Perception Scale Item Statistics (V1, Artificial Agent)

Item Code	Piers				RL			
	Item <i>M</i>	Item <i>SD</i>	Confidence <i>M</i>	Confidence <i>SD</i>	Item <i>M</i>	Item <i>SD</i>	Confidence <i>M</i>	Confidence <i>SD</i>
TPA01	4.00	0.93	6.38	1.51	2.88	0.99	6.75	1.39
TPA02	4.13	0.84	6.13	1.46	3.13	1.36	6.63	1.30
TPA03	4.00	1.07	6.00	1.20	2.75	1.04	6.38	1.19
TPA04	5.00	0.93	7.63	1.77	4.50	1.41	7.75	1.83
TPA05	4.50	1.20	6.88	2.10	4.38	1.06	7.63	1.60
TPA06	5.13	0.84	7.88	1.96	4.88	0.84	7.38	1.92
TPA07	4.50	0.93	6.63	3.20	2.88	1.25	6.88	1.13
TPA08	2.63	0.92	5.50	1.77	2.25	0.89	5.88	1.89
TPA09	3.88	1.13	6.00	2.51	2.75	0.89	6.38	1.85